Mechanical Ventilation, Adult

AR-11 | Airway & Respiratory

Quick flow

01 Advanced Airway Confirmed [Paramedic]

Continuous waveform EtCO2 - tube depth - bilateral exam - securement - analgesia and sedation through AR-08.

02 Functioning Home / Transfer Ventilator? [Paramedic]

Maintain established settings when safe; verify baseline, reserve oxygen/power, BVM, suction, and caregiver plan.

03 New Controlled Setup [Paramedic]

Volume assist-control; tidal volume 6-8 mL/kg IBW. Default to 6 mL/kg for ARDS/restrictive physiology or elevated pressures.

04 Choose Rate by Physiology [Paramedic]

Usual adult: 10-12/min. COPD/asthma: 8-10/min, high inspiratory flow, and prolonged exhalation.

05 Oxygenation + Pressure Check [Paramedic]

PEEP 5 - FiO2 100% only until reliable oximetry - SpO2 goal 92-98% - assess synchrony, exhalation, and plateau pressure.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Alarm / Deterioration? [EMT / AEMT / Paramedic]

If cause is not immediately clear: disconnect, BVM, evaluate DOPES/DOTT, correct cause, and notify destination.

Overview

- For a new controlled ventilator setup, use lung-protective volume assist-control ventilation based on Ideal Body Weight (IBW) from height and sex, not actual scale weight.
- Maintain stable, functioning established home or transfer settings unless the patient's clinical condition requires change. If an alarm occurs and the cause is uncertain, disconnect immediately and manually ventilate while evaluating the cause.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Continuously monitor the airway, SpO_2, patient color, chest movement, and ventilator connection; manually ventilate immediately if failure occurs.
- Report alarms, loss of waveform, increased resistance, or clinical deterioration.

AEMT

- Perform all EMT care plus capnography, authorized airway care, suctioning, and assistance with ventilator troubleshooting.

Paramedic

- For a new controlled setup, use volume assist-control and calculate tidal volume from Ideal Body Weight (IBW) using height and sex; do not use actual scale weight.
- Set tidal volume at 6-8 mL/kg IBW. Start at 6 mL/kg for ARDS, restrictive physiology, or elevated pressures, and only increase when clinically appropriate while maintaining plateau pressure below 30 cm H2O.
- For usual adult physiology, begin 10-12/min and reassess ventilation from waveform EtCO2 trend, pH/PaCO2 information when available, chest excursion, synchrony, and clinical response. For COPD/asthma, begin 8-10/min with high inspiratory flow and I:E about 1:4-1:5; ensure expiratory flow returns to baseline.
- Measure plateau pressure only during an inspiratory hold in a passive patient. For plateau pressure above 30 cm H2O, reduce tidal volume in 1 mL/kg IBW steps to no lower than 4 mL/kg IBW while correcting the underlying cause and obtaining additional direction as needed.

Indications

- Adult or protocol-authorized patient with advanced airway requiring mechanical ventilation.
- Interfacility or home-ventilator patient whose current settings are known and functioning.

Treatment steps

- Initial new controlled setup: volume assist-control, tidal volume 6-8 mL/kg IBW, PEEP 5 cm H2O, and FiO2 100% only until a reliable oxygen-saturation measurement is available. Then use the lowest effective FiO2 and small, reassessed PEEP changes to maintain SpO2 92-98% unless a condition-specific target applies.
- After return of spontaneous circulation, use the post-resuscitation target SpO2 90-98% and avoid using EtCO2 as a direct substitute for PaCO2. When blood-gas data are available, target PaCO2 35-45 mm Hg unless directed otherwise.
- For suspected traumatic brain injury, follow the TB-05 targets below rather than an unspecified normal-ventilation goal. In severe metabolic acidosis, do not abruptly remove compensatory minute ventilation; reassess closely and seek Medical Control or receiving-facility direction.
- For high PIP with high plateau pressure, suspect reduced compliance (lung, chest wall, or abdominal); for high PIP with normal plateau pressure, suspect airway/circuit resistance or auto-PEEP.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Warnings / precautions

- Tidal volume is based on Ideal Body Weight (IBW) from height and sex, never actual body weight.
- Do not persist with an alarming ventilator when the cause is unclear; disconnect and manually ventilate. In obstructive lung disease with suspected auto-PEEP, allow full exhalation and ventilate slowly after disconnecting.
- Do not force EtCO2 to a numeric target in severe obstructive lung disease when it worsens air trapping. EtCO2 may not equal PaCO2, especially after ROSC or poor perfusion.
- Do not use this new-setup pathway to overwrite stable home or transfer settings without a clinical reason.

Clinical pearls

- DOPES: displacement, obstruction, pneumothorax/pulmonary problem, equipment failure, stacked breaths/auto-PEEP.
- DOTT response: disconnect or decompress when indicated, oxygen/BVM, tube check, then adjust settings only after the cause is addressed.
- Document height, sex used for IBW, all initial and changed settings, oxygen/EtCO2 trends, pressure values, waveform findings, sedation/analgesia, alarms, troubleshooting, and patient response.

Related pathways / contacts

AR-08 Post-Intubation / BIAD Management - Open for analgesia, sedation, continuous airway confirmation, and post-airway reassessment.

AR-09 Ventilator Emergencies - Open for home-ventilator failure, equipment troubleshooting, transport preparation, and DOPES response.

AC-10 Post-Resuscitation Care - Open after ROSC for oxygenation, ventilation, blood-pressure, and destination priorities.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS medical-director approved adult mechanical-ventilation pathway, October 17, 2026. | American Thoracic Society clinical practice guideline for adult ARDS ventilation. | American Heart Association post-cardiac-arrest care guidance.

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